

$$H = \frac{p^2}{2m} + \frac{1}{2} k x^2 = \hbar \omega (a^\dagger a + \frac{1}{2}), \quad \omega = \sqrt{\frac{k}{m}}$$

$$a = \frac{1}{\sqrt{2}} \left(\frac{x}{x_0} + i \frac{p}{p_0} \right), \quad x_0 = \sqrt{\frac{\hbar}{m\omega}}, \quad p_0 = \frac{\hbar}{x_0}$$

$$x = \frac{x_0}{\sqrt{2}} (a + a^\dagger) \quad \frac{d}{dt} \langle A, t \rangle = \frac{i}{\hbar} [H, A], t$$

$$p = \frac{p_0}{i\sqrt{2}} (a - a^\dagger) \quad \frac{d}{dt} \langle x, t \rangle = \frac{\langle p, t \rangle}{m}$$

$$[a, a^\dagger] = 1 \quad \frac{d}{dt} \langle p, t \rangle = -k \langle x, t \rangle$$

$$H|n\rangle = \hbar \omega (n + \frac{1}{2}) |n\rangle, \quad n = 0, 1, 2, \dots$$

$$a|n\rangle = \sqrt{n} |n-1\rangle \quad a(t) = a e^{-i\omega t}$$

$$a^\dagger|n\rangle = \sqrt{n+1} |n+1\rangle \quad \langle n|m\rangle = \delta_{nm}$$

$$a^\dagger a |n\rangle = n |n\rangle$$

$$\textcircled{1} |\psi, 0\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{i}{\sqrt{2}} |1\rangle$$

$$\langle x, t \rangle = ?$$

$$\langle p, t \rangle = ?$$

$$\delta x(t) = ?$$

$$\delta p(t) = ?$$

A) Schrödingerjeva slika

$$|\psi, t\rangle = \frac{1}{\sqrt{2}} e^{i \frac{\hbar \omega}{2} t} |0\rangle + \frac{i}{\sqrt{2}} e^{-i \frac{3\hbar \omega}{2} t} |1\rangle$$

$$\langle x, t \rangle = \langle \psi, t | x | \psi, t \rangle = \frac{x_0}{\sqrt{2}} \langle \psi, t | a + a^\dagger | \psi, t \rangle = \frac{2x_0}{\sqrt{2}} \langle \psi, t | \text{Re } a | \psi, t \rangle =$$

$$= \sqrt{2} x_0 \text{Re} \langle a, t \rangle$$

$$\langle p, t \rangle = \langle \psi, t | p | \psi, t \rangle = \frac{p_0}{\sqrt{2}i} \langle \psi, t | a - a^\dagger | \psi, t \rangle = \frac{2ip_0}{\sqrt{2}i} \langle \psi, t | \text{Im } a | \psi, t \rangle =$$

$$= \sqrt{2} p_0 \text{Im} \langle a, t \rangle$$

$$\langle a, t \rangle = \langle \psi, t | a | \psi, t \rangle = \langle \psi, t | \left(\frac{1}{\sqrt{2}} e^{i \frac{\omega t}{2}} \overset{0}{a} |0\rangle + \frac{i}{\sqrt{2}} e^{\frac{3i\omega t}{2}} \overset{\sqrt{1}|0\rangle}{a} |1\rangle \right) =$$

$$= \left(\frac{1}{\sqrt{2}} e^{i\frac{\omega t}{2}} \langle 0| - \frac{i}{\sqrt{2}} e^{i\frac{3}{2}\omega t} \langle 1| \right) \left(\frac{i}{\sqrt{2}} e^{-i\frac{3}{2}\omega t} |0\rangle \right) =$$

$$= \frac{i}{2} e^{i\frac{\omega t}{2}} e^{-\frac{3i\omega t}{2}} = \frac{i}{2} e^{-i\omega t} = \frac{i}{2} [\cos \omega t - i \sin \omega t] = \boxed{\frac{i}{2} \cos \omega t + \frac{1}{2} \sin \omega t}$$

$$\rightarrow \boxed{\langle x, t \rangle = \sqrt{2} x_0 \frac{1}{2} \sin \omega t}$$

$$\boxed{\langle p, t \rangle = \sqrt{2} p_0 \frac{1}{2} \cos \omega t}$$

B) Heisenbergova slika

$$\langle a, t \rangle = \langle a(t), 0 \rangle ; \quad \frac{d}{dt} a(t) = \frac{i}{\hbar} [H, a](t) = -\frac{i}{\hbar} \hbar \omega a(t) = -i\omega a(t) ; \quad \text{zP } a(0) = a$$

$$[H, a] = \hbar \omega \left[a^\dagger a + \frac{1}{2}, a \right] = \hbar \omega \left(a^\dagger \underbrace{[a, a]}_{-1} + [a^\dagger, a] a \right) = -\hbar \omega a$$

$$\rightarrow a(t) = A e^{-i\omega t} \rightarrow a(0) = A = a \Rightarrow a(t) = a e^{-i\omega t}$$

$$\begin{aligned} \langle x, t \rangle &= \sqrt{2} x_0 \operatorname{Re} \langle a, t \rangle = \sqrt{2} x_0 \operatorname{Re} \langle a(t), 0 \rangle = \sqrt{2} x_0 \operatorname{Re} \langle a e^{-i\omega t}, 0 \rangle = \sqrt{2} x_0 \operatorname{Re} \left(e^{-i\omega t} \underbrace{\langle a, 0 \rangle}_{\frac{1}{2}} \right) = \\ &= \dots \text{ (isto kot pri A)} \end{aligned}$$

$$\langle p, t \rangle = \dots \text{ (podobno)}$$

$$\delta^2 x(t) = \langle x^2, t \rangle - \langle x, t \rangle^2$$

$$[a, a^\dagger] = 1 = a a^\dagger - a^\dagger a \Rightarrow a a^\dagger = 1 + a^\dagger a$$

$$\delta^2 p(t) = \langle p^2, t \rangle - \langle p, t \rangle^2$$

$$\begin{aligned} \langle x^2, t \rangle &= \frac{x_0^2}{2} \langle (a + a^\dagger)^2, t \rangle = \frac{x_0^2}{2} \left[\langle a^2, t \rangle + \langle a a^\dagger + a^\dagger a, t \rangle + \langle a^{\dagger 2}, t \rangle \right] = \\ &= \frac{x_0^2}{2} \left[2 \operatorname{Re} \langle a^2, t \rangle + 2 \langle a^\dagger a, t \rangle + 1 \right] = x_0^2 \left[\operatorname{Re} \langle a^2(t), 0 \rangle + \langle a^\dagger a(t), 0 \rangle + \frac{1}{2} \right] \end{aligned}$$

$$\rightarrow a^2(t) = a(t)^2 = a^2 e^{-2i\omega t}$$

$$a^\dagger a(t) = a^\dagger(t) a(t) = a(t)^\dagger a(t) = a^\dagger e^{i\omega t} a e^{-i\omega t} = a^\dagger a$$

$$\rightarrow \langle x^2, t \rangle = x_0^2 \left[\operatorname{Re} \left(e^{-2i\omega t} \underbrace{\langle a^2, 0 \rangle}_0 \right) + \underbrace{\langle a^\dagger a, 0 \rangle}_{\frac{1}{2}} + \frac{1}{2} \right] = x_0^2$$

$$\langle p^2, t \rangle = \dots = p_0^2$$

podobno

$$\rightarrow \delta^2 x(t) = x_0^2 \left(1 - \frac{1}{2} \sin^2 \omega t \right)$$

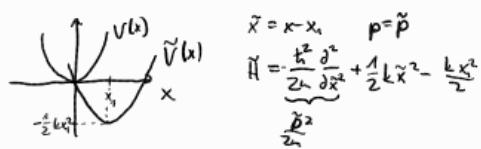
$$\delta^2 p(t) = p_0^2 \left(1 - \frac{1}{2} \cos^2 \omega t \right)$$

LHO, ob t=0 vključimo el. polje

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = i\hbar \frac{\partial \Psi}{\partial t}, \quad V(x) = \frac{1}{2} kx^2$$

ZP: $\langle x \rangle = 0, \quad \langle p \rangle = 0$
 $\delta x = m\sigma, \quad \delta p = m\sigma$
 $\delta x \delta p = \frac{\hbar}{2} \Rightarrow$ Gaussov paket: $\Psi(x,0) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-x_0)^2}{4\sigma^2}} e^{i\frac{p_0}{\hbar}x} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{4\sigma^2}}$
 $\delta x = \sigma, \quad \delta p = \frac{\hbar}{2\sigma}$
 želimo se E minimalna $\Rightarrow E = \frac{\hbar\omega}{2}$ osh. stanje $\Rightarrow \sigma^2 = \frac{\hbar^2}{2k}$

$\langle x, t \rangle = ?$
 $\langle p, t \rangle = ?$
 $\delta x(t) = ?$
 $\delta p(t) = ?$
 $t=0: H = \frac{p^2}{2m} + \frac{1}{2} kx^2$
 $t>0: H = \frac{p^2}{2m} + \frac{1}{2} kx^2 - e\epsilon x = \frac{p^2}{2m} + \frac{1}{2} k(x - \frac{e\epsilon}{k})^2 - \frac{(e\epsilon)^2}{2k}$
 $= \frac{p^2}{2m} + \frac{1}{2} k(x - \frac{e\epsilon}{k})^2 - \frac{(e\epsilon)^2}{2k} = \frac{p^2}{2m} + \frac{1}{2} k(x - x_1)^2 - x_1^2 \frac{k}{2} = \frac{p^2}{2m} + \tilde{V}(x)$



$\rightarrow a = \frac{1}{\sqrt{2}} \left(\frac{x}{x_0} + i \frac{p}{p_0} \right)$
 $\tilde{a} = \frac{1}{\sqrt{2}} \left(\frac{\tilde{x}}{x_0} + i \frac{\tilde{p}}{p_0} \right) = a - \frac{1}{\sqrt{2}} \frac{x_1}{x_0} \rightarrow \tilde{a}|\psi_0\rangle = -\frac{1}{\sqrt{2}} \frac{x_1}{x_0} |\psi_0\rangle$
 $\tilde{a}|\psi\rangle = z|\psi\rangle \quad z \in \mathbb{C}$

stranski razum: koherentna stanja

$H = \frac{p^2}{2m} + \frac{1}{2} kx^2, \quad a = \frac{1}{\sqrt{2}} \left(\frac{x}{x_0} + i \frac{p}{p_0} \right)$
 $a|\psi\rangle = z|\psi\rangle, \quad z \in \mathbb{C}$
 $\downarrow$
 $|\psi\rangle = \sum c_n |n\rangle$
 $a|\psi\rangle = z \sum c_n |n\rangle = \sum z c_n |n\rangle$
 $a \sum c_n |n\rangle = \sum c_n \sqrt{n} |n-1\rangle = \sum c_{n+1} \sqrt{n+1} |n\rangle$
 $\rightarrow z c_n = \sqrt{n+1} c_{n+1} \rightarrow c_{n+1} = \frac{z}{\sqrt{n+1}} c_n \rightarrow c_n = \frac{z^n}{n!} c_0$

$\rightarrow |\psi\rangle = \sum_{n=0}^{\infty} \frac{z^n}{n!} c_0 |n\rangle$
 $\langle \psi | \psi \rangle = 1 = \left(\sum \frac{z^{n*}}{n!} c_0^* \langle n| \right) \left(\sum \frac{z^n}{n!} c_0 |n\rangle \right) = \sum_{n,n'} \frac{z^{n*}}{n!} |c_0|^2 \frac{z^{n'}}{n!} \langle n | n' \rangle = \sum_n \frac{|z^n|^2}{n!} |c_0|^2 = e^{|z|^2} |c_0|^2 = 1 \rightarrow c_0 = e^{-\frac{1}{2}|z|^2}$

$\rightarrow |\psi\rangle = e^{-\frac{1}{2}|z|^2} \sum \frac{z^n}{n!} |n\rangle$
 $|\psi, t\rangle = e^{-\frac{1}{2}|z|^2} \sum \frac{z^n}{n!} e^{-i\frac{E_n}{\hbar}t} |n\rangle = e^{-\frac{1}{2}|z|^2} \sum \frac{z^n}{n!} e^{-i\omega(n+\frac{1}{2})t} |n\rangle = e^{-\frac{1}{2}|z|^2 - \frac{1}{2}i\omega t} \sum \frac{(ze^{-i\omega t})^n}{n!} |n\rangle = \frac{e^{-i\frac{\omega}{2}t} e^{-\frac{1}{2}|ze^{-i\omega t}|^2} \sum \frac{(ze^{-i\omega t})^n}{n!} |n\rangle}{z \rightarrow ze^{-i\omega t}}$

$a|\psi\rangle = z|\psi\rangle$
 $\downarrow$
 $a|z\rangle = z|z\rangle$
 koherentna stanja
 $|z, t\rangle = e^{-i\frac{\omega}{2}t} |ze^{-i\omega t}\rangle$

$\langle x \rangle = \langle z | x | z \rangle = \sqrt{2} x_0 \text{Re}(\langle z | a | z \rangle) = \sqrt{2} x_0 \text{Re}(\underbrace{\langle z | z | z \rangle}_{\text{vektor}}) = \sqrt{2} x_0 \text{Re}(\underbrace{\langle z | z | z \rangle}_1) = \sqrt{2} x_0 \text{Re}(z)$
 $\langle p \rangle = \langle z | p | z \rangle = \sqrt{2} p_0 \text{Im}(\langle z | a | z \rangle) = \sqrt{2} p_0 \text{Im}(z)$
 NORMALNI VREDNI R.E.D
 $\langle z | a^{\dagger n} a^n | z \rangle = \langle z | a^{\dagger n} z^n | z \rangle = z^n \langle z | a^{\dagger n} | z \rangle = z^n \langle a^{\dagger n} z | z \rangle = z^n \langle z^{\dagger n} z | z \rangle = z^n (z^{\dagger n})$

! $\langle z | a a^{\dagger} | z \rangle = \langle z | 1 + a^{\dagger} a | z \rangle = 1 + z^* z$
 $\langle x^2 \rangle = x_0^2 (\text{Re} \langle a^2 \rangle + \langle a^{\dagger} a \rangle + \frac{1}{2}) = x_0^2 (\text{Re}(z^2) + z^* z + \frac{1}{2})$
 $\langle p^2 \rangle = p_0^2 (-\text{Re} \langle a^2 \rangle + \langle a^{\dagger} a \rangle + \frac{1}{2}) = p_0^2 (-\text{Re}(z^2) + z^* z + \frac{1}{2})$

$$\delta x^2 = \langle x^2 \rangle - \langle x \rangle^2 = x_0^2 \left(\text{Re}(z^2) + z^* z + \frac{1}{2} \right) - 2x_0^2 \text{Re}(z) = x_0^2 \left(\frac{1}{2}(z^2 + z^{*2}) + z^* z + \frac{1}{2}(z^2 + 2zz^* + z^{*2}) \right) = \frac{1}{2}x_0^2$$

$$\downarrow \quad \downarrow$$

$$\text{Re}(z) = \frac{z^2 + z^{*2}}{2} \quad \text{Re}(z) = \frac{z + z^*}{2}$$

$$\delta^2 p = \dots = \frac{1}{2} p_0^2$$

$$\rightarrow \delta x \delta p = \frac{1}{2} x_0 p_0 = \frac{1}{2} \hbar \quad \checkmark$$

$$\langle H \rangle = \hbar \omega \left(a^\dagger a + \frac{1}{2} \right) = \hbar \omega \left(z^* z + \frac{1}{2} \right)$$

$$\langle H^2 \rangle = (\hbar \omega)^2 \left(\underbrace{a^\dagger a a^\dagger a}_{a^\dagger (1 + a^\dagger a) a} + \frac{1}{2} a^\dagger a + \frac{1}{2} a^\dagger a + \frac{1}{4} \right) = (\hbar \omega)^2 \left(a^{\dagger 2} a^2 + 2a^\dagger a + \frac{1}{4} \right) = (\hbar \omega)^2 \left(z^{*2} z^2 + 2z^* z + \frac{1}{4} \right)$$

$$\rightarrow \delta^2 H = (\hbar \omega)^2 (z^* z)$$

$$\langle \psi, 0 \rangle = 0$$

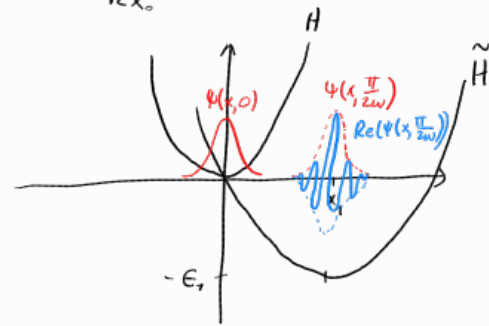
$$z | \psi, 0 \rangle = -\frac{x_1}{\sqrt{2} x_0} | \psi, 0 \rangle$$

$$\Rightarrow z = -\frac{x_1}{\sqrt{2} x_0} \in \mathbb{R}$$

$$| \psi, t \rangle = e^{-\frac{i \omega t}{2}} | z e^{-i \omega t} \rangle \cdot e^{-i \frac{\epsilon_0}{\hbar} t}$$

$$\langle x, t \rangle = \langle x, t \rangle + x_1 = \sqrt{2} x_0 \text{Re}(z e^{-i \omega t}) + x_1 =$$

$$= \sqrt{2} x_0 \left(-\frac{x_1}{\sqrt{2} x_0} \right) \text{Re}(e^{-i \omega t}) + x_1 = \boxed{x_1 (1 - \cos \omega t)}$$



▼ širina val. paketa = konst.

2D:

$$H = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{1}{2} k_x x^2 + \frac{1}{2} k_y y^2 = H_x + H_y$$

$$H_x \psi_x = E_x \psi_x(x)$$

$$H_y \psi_y = E_y \psi_y(y)$$

$$H \psi = E \psi(x, y)$$

$$\rightarrow \psi = \psi_x \psi_y \quad \begin{matrix} \nearrow \sqrt{\frac{k_x}{m}} \\ \nearrow \sqrt{\frac{k_y}{m}} \end{matrix}$$

$$E = E_x + E_y = \hbar \omega_x \left(n_x + \frac{1}{2} \right) + \hbar \omega_y \left(n_y + \frac{1}{2} \right)$$

$$H |n_x, n_y\rangle = E |n_x, n_y\rangle \equiv H |n_x, n_y\rangle = E |n_x, n_y\rangle$$

$$\textcircled{1} k_x > 0, k_y = 0$$

$$E = \hbar \omega_x \left(n_x + \frac{1}{2} \right) + \frac{\hbar^2 q^2}{2m}$$

$$\psi_y = e^{iqy}$$

$$\textcircled{2} k_x = k_y = k$$

$$E = \hbar \omega (n_x + n_y + 1)$$

$$H = \frac{p^2}{2m} + \frac{1}{2} k r^2$$

